Eigenvalues And Eigenvectors

Introduction

Imagine you're developing a software to apply transformations — like scaling, rotation, and distortion — to 2D game characters or 3D objects. Some directions change wildly… but surprisingly, there are certain 'magic directions' that keep their orientation when transformed — they just stretch or shrink.

These are called **Eigenvectors**.

And how much they stretch or shrink is the **Eigenvalue**.

What are Eigenvectors and Eigenvalues

Let's break it down with simple Linear Algebra:

Given a **square matrix** $\boldsymbol{A}$, an **eigenvector** $\vec{\boldsymbol{v}}$ satisfies $A\vec{\boldsymbol{v}} = \boldsymbol{\lambda}\vec{\boldsymbol{v}}$

Here

$\vec{\boldsymbol{v}}$: a **non-zero vector** (direction that doesn't rotate)

$\boldsymbol{\lambda}$: the **eigenvalue** (scalar — tells how much it scales)

$\boldsymbol{A}$: a **transformation matrix** (e.g., scaling, rotation, projection)

In short: **An eigenvector points in a direction that stays unchanged except for scaling when a matrix is applied.**

Video Recommendation System

When you watch a tech video, the system projects your activity into a “preference space.” This space has directions like “tech-lover,” “vlogger-fan,” “gamer,” etc. These directions are learned using eigenvector-type methods.
Your behavior vector aligns most with, say, the “tech” direction. Then the algorithm finds **other videos aligned along that direction** — just like projecting along eigenvectors.

Concept	YouTube Analogy
Matrix	User–video interaction matrix
Eigenvector	Direction capturing a viewing pattern
Eigenvalue	Strength/importance of that pattern
SVD/PCA	Techniques used to reduce noise and find dominant patterns
Recommendation	Videos aligned with your strongest behavior "direction"

Example 1

Eigenvector of a 2×2 Matrix

Let us check that vector $\boldsymbol{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigenvector of

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

corresponding to the eigenvalue $\lambda = 3$.

Since

$$A * \boldsymbol{x} = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} = 3\boldsymbol{x}$$

Hence the result.

Theorem

If $\boldsymbol{A}$ is $n \times n$ matrix, then $\boldsymbol{\lambda}$ is an eigenvalue of **A** if and only if it satisfies the equation

$$det(A - \lambda I) = 0 \tag{1}$$

This is called the **characteristic equation** of **A**.

Note: In Example 1 we observed that $\lambda = 3$ is an eigenvalue of the matrix

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

but we did not explain how we found it. Use the characteristic equation to find all eigenvalues of this matrix.

Example 2: Find eigenvalues of the matrix A $\begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$

Solution: The eigenvalues of A are the solution of the equation $det(A-\lambda I)=0$. As

$$A-\lambda I=\begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}-\lambda\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A-\lambda I=\begin{bmatrix} 3-\lambda & 0 \\ 8 & -1-\lambda \end{bmatrix}$$

$$\Rightarrow det(A-\lambda I)=\begin{vmatrix} 3-\lambda & 0 \\ 8 & -1-\lambda \end{vmatrix}$$

Therefore, the characteristic equation $det(A-\lambda I)=0$ is

$$\begin{vmatrix} 3-\lambda & 0 \\ 8 & -1-\lambda \end{vmatrix}=0$$

$$\Rightarrow \quad (3-\lambda)(-1-\lambda)-8(0)=0$$

$$\Rightarrow -(3-\lambda)(1+\lambda)=0$$

So, eigenvalues are $\lambda=3,\ \lambda=-1$

Theorem

If A is an n × n triangular matrix (upper triangular, lower triangular, or diagonal), then the eigenvalues of A are the entries on the main diagonal of A.

Example | Eigen values of Lower Triangular Matrix

By inspection, the eigenvalues of the lower triangular matrix

$$A = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ -1 & \frac{2}{3} & 0 \\ 5 & -8 & -\frac{1}{4} \end{bmatrix}$$

are $\lambda = \frac{1}{2}$, $\lambda = \frac{2}{3}$, and $\lambda = -\frac{1}{4}$.

Please see the following MS Word file uploaded on portal for further practice problems

17. Eigenvalues and Eigenvectors

THANKS